

# Module 9: Fundamental Concepts of Loops

## Practice Task

### Task 1 — Print "Hello" Five Times

Write a for loop that prints "Hello" 5 times.

```
// Your code here
```

### Task 2 — Print Numbers 1 to 10

Write a for loop that prints numbers from 1 to 10.

```
// Your code here
```

### Task 3 — Countdown with While Loop

Write a while loop that prints numbers from 10 down to 1.

```
// Your code here
```

### Task 4 — Repeat a Message

Using a while loop, print "Loop চলছে" exactly 7 times.

```
// Your code here
```

### Task 5 — Even Numbers

Write a for loop that prints all even numbers between 1 and 30.

```
// Your code here
```

### Task 6 — Sum of Numbers

Write a for loop that calculates the sum of numbers from 1 to 20 and prints the result.

```
// Your code here
```

### Task 7 — Multiplication Table

Ask the user for a number (or set a variable), then print its multiplication table from 1 to 10 using a for loop. Example: if the number is 7 →  $7 \times 1 = 7$ ,  $7 \times 2 = 14$ , ...  $7 \times 10 = 70$ .

```
// Your code here
```

### Task 8 — Count Down from 20

Write a for loop that counts down from 20 to 1.

```
// Your code here
```

### Task 9 — Sum of Even Numbers

Write a for loop that calculates the sum of all even numbers from 2 to 50.

```
// Your code here
```

### Task 10 — Using Break

Write a for loop from 1 to 50 that stops completely (break) as soon as it reaches a number greater than 30.

```
// Your code here
```

### Task 11 — Using Continue

Write a for loop from 1 to 20 that skips (continue) any number divisible by 4.

```
// Your code here
```

## Task 12 — Do While Loop — Optional

Write a do while loop that prints numbers from 1 to 5. Then modify the condition so it is false from the start (for example,  $i > 10$ ) and observe that the loop still runs once.

```
// Your code here
```

## Task 13 — Comparing Loop Types

Write the same task — printing numbers 1 to 5 — three times: once using for, once using while, and once using do while. Compare how the code structure differs for each.

```
// for loop version
```

```
// while loop version
```

```
// do while loop version
```

## Task 14 — Mini Challenge

Write a program that:

1. Loops from 1 to 100.
2. Skips (continue) any number divisible by 5.
3. Stops (break) completely once it reaches a number greater than 40.
4. Prints every number that passes both checks.

```
// Your code here
```

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